

```

1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 2: pure python
5
6  @author Austin Downey
7  """
8
9  # clears everything, including IPython
10 import IPython as IP
11 IP.get_ipython().run_line_magic('reset', '-sf')
12
13 # import modules native to Python, use the "as" to not get confused
14 import random as random # random number generator
15 import sys as sys # import systems functions
16 import os as os # import os functions
17
18 # <-- makes a comment
19
20 ''' <-- makes a multi
21 line
22 comment '''
23
24 %% makes a section, means nothing in the code, helpful for splitting up code
25
26
27 print('hello world') # prints a string to the console.
28 print("hello world") # prints the same to the console, but you have to hit the shift
29 print(' "hello world" ' ) # allows you to print the quotation marks to the console.
30
31 # print will retrun a string, form anything it can.
32 print('string')
33 print(10) # number
34 print([10]) # string
35 # but it cant print two typtes of variables unless seperated by a ,
36 print('string',10,[10])
37 # or concatated into a string
38 print('string '+ str(10)+ ' ' + str([10]))
39
40
41
42 %% arithmetic operations
43 # There are seven different arithmetic operations
44 '''
45 + addition - yields the sum of its arguments
46 - subtraction - yields the difference of its arguments
47 * multiplication - yields the product of its arguments
48 / division - yields the quotient of its arguments
49 % modulo - yields the remainder from the division of the first argument by the second.
50 ** exponential - yields the power (or exponentiation) of its arguments
51 // floor division - yield the quotient of its arguments with the 'floor' function
52 applied to the result
53 '''
54
55 print('5 + 2 = ', 5+2)
56 print('5 - 2 = ', 5-2)
57 print('5 * 2 = ', 5*2)
58 print('5 / 2 = ', 5/2)
59 print('5 % 2 = ', 5%2)
60 print('5 ** 2 = ', 5**2)
61 print('5 +// 2 = ', 5//2)
62
63 # order of operation matters in Python,
64 print('1 + 2 - 3 * 2 = ', 1 + 2 - 3 * 2 )
65 print('(1 + 2 - 3) * 2 = ', (1 + 2 - 3) * 2 )
66

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67  %% variables.
68
69  # variables have to start with a number and can have numbers and underscores
70
71  name = 'Austin'
72  last_name = 'Downey'
73
74  ''' there are lots of data types (classes) in Python, we will use seven main ones.
75  integers (class int)
76  floats (class float)
77  booleans (class bool)
78  strings (class str)
79  lists (class list)
80  tuples (class tuple)
81  dictionaries (class dict)
82  '''
83
84  ii = 10
85  ff = 10/2
86  bb = False
87  ss = 'Austin'
88  ll = [1,2,'3',ss,bb] # a list of anything.
89  # a list will copy, print list, change bb to True, reprint list.
90  tt = (6,7,8,9,10) # a list of anything that you can not change.
91  dd = {'ii':ii,
92        'ff':ff,
93        'bb':bb,
94        'ss':ss,
95        'll':ll,
96        'tt':tt}
97
98  %% control flow statements.
99
100  ''' The Python for statement iterates over the members of a sequence in order,
101  executing the block each time. '''
102
103  for i in range (0,3):
104      print("We're on time %d" % (i))
105
106  ''' the 'while' loop, used when a condition needs to be checked each iteration
107  or to repeat a block of code forever. '''
108
109  x = 1
110  while x<100: #True:
111      print("To infinity and beyond! We're getting close, on %d now!" % (x))
112      x += 1
113
114
115  ''' The while loop and the for loop can be made to exit before the given object
116  is finished. This is done using the break statement, which will immediately drop
117  out of the loop and continue execution at the first statement after the block. '''
118
119  for x in range(3):
120      if x == 1:
121          break
122
123  ''' You can also have an optional else clause, which will run should the for loop
124  exit cleanly - that is, without breaking. '''
125
126  for x in range(3):
127      print(x)
128  else:
129      print('Final x = %d' % (x))
130
131
132  ''' Lastly, you can directly use an "if statement", which will allow you to cycle
133  through statements to check. '''

```

```
134
135 i='d'
136 if i == 'a':
137     print('the variable is ' + i)
138 elif i == 'b':
139     print('the variable is ' + i)
140 elif i == 'c':
141     print('the variable is ' + i)
142 elif i == 'd':
143     print('the variable is ' + i)
144 else:
145     print('the variable is a mystery')
146
```